EBRAINS Data Descriptor

TITLE*.

Probabilistic cytoarchitectonic map of the Area TPJ (STG/SMG) (v6.2)

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ABSTRACT*

This dataset contains cytoarchitectonic probability maps of the Area TPJ (STG/SMG) in the single subject template of the MNI ("Colin 27") and the MNI ICBM 152 reference space, as part of the Julich-Brain atlas. The area was identified using cytoarchitectonic analysis on cell-body-stained histological sections in both hemispheres of 10 human postmortem brains obtained from the body donor program of the University of Düsseldorf. The maps show the probability to belong to the Area TPJ (STG/SMG) for each voxel of the reference spaces. Data is provided in the NifTi format. The Julich-Brain atlas relies on a modular, flexible and adaptive framework containing workflows to create probabilistic brain maps. Note that methodological improvements and integration of new brain structures may lead to small deviations in earlier released datasets.

Delineations of the Area TPJ (STG/SMG) derived from the calculation of a maximum probability map of all currently released Julich-Brain brain structures can be found here:

Amunts et al. (2021) [Data set, v2.9]

METHODS*

The methods followed the procedure as described in Amunts et al. (2020). Cytoarchitectonic maps were generated based on the analysis of histological sections from ten human brains (5 female, 5 male). The brains were obtained in accordance to legal and ethical regulations and guidelines as part of the body donor program of the Department of Anatomy of the Heinrich Heine University Düsseldorf. Written informed consent for the general use of postmortem tissue for aims of research and education was obtained from the body donors. All usage in this work is covered by a vote of the ethics committee of the Medical Faculty of the Heinrich Heine University Düsseldorf (#4863). The brains were fixed in 4% buffered formalin or Bodian's fixative for at least 6 months. All brains underwent magnetic resonance imaging (MRI) using a T1-weighted 3D FLASH sequence (flip angle 40°, repetition time TR 40 ms, echo time TE 5 ms). MRI images were used as an undistorted spatial reference for the 3D-reconstruction of the

histological sections. After scanning, the brains were embedded in paraffin and serially sectioned in the coronal plane on a large-scale microtome (20 µm thickness). Every 15th section was stained for cell bodies using a silver staining technique.
Subsequently, the Area TPJ (STG/SMG) was delineated and traced on every 30th section of each brain on high-resolution digitized scans of the sections (1 µm in-plane resolution) using cytoarchitectonic characteristics. The images of histological sections of each brain and the corresponding delineations of the Area TPJ (STG/SMG) were subsequently 3D-reconstructed using the MRI data sets of the fixed brains and the high-resolution digitized scans of the sections. In a next step the delineations of all examined brains were spatially normalized and transferred to the Montreal Neurological Institute (MNI) reference spaces MNI Colin 27 and MNI ICBM 152 (Evans et al., 2012) to calculate probabilistic maps. Values from 0% to 100% indicate the probability for each voxel of the reference brains to contain the Area TPJ (STG/SMG) at a certain position.

TECHNICAL VALIDATION*

The delineations of the Area TPJ (STG/SMG) were compared across all ten brains by neuroanatomical experts to verify anatomical plausibility with regards to topography and neighboring anatomical structures.

USAGE NOTES*

As part of the Julich-Brain atlas, the probabilistic Area TPJ (STG/SMG) maps allow for a comparison of functional activations, networks, genetic expression patterns, anatomical structures, and other data obtained across different studies in a common stereotaxic reference space (Amunts et al., 2020).

SPATIAL ANCHORING:

The generated cytoarchitectonic map of the Area TPJ (STG/SMG) has been integrated into the HBP Human Brain Atlas. It was spatially anchored to the MNI Colin 27 and MNI ICBM152 2009c nonlinear asymmetric reference spaces (Evans et al., 2012) and is accessible as part of the Julich-Brain atlas (Amunts et al., 2020). Delineations of the Area TPJ (STG/SMG) derived from the calculation of a maximum probability map of all currently released Julich-Brain brain structures can be found here: Amunts et al. (2021) [Data set, v2.9]

DATA RECORDS*

In the repository the data are stored in the following structure (incl. info on file content):

hbp-d000001_jubrain-cytoatlas_dev/
- **PMs/**
  - **Area-TPJ/**
    - **6.2/**
      - **EBRIANS-DataDescriptor_TPJ_CytoMap.pdf** [contains a short description of the dataset]
      - **Licence-CC-BY-NC-SA.pdf**
      - **subjects_Area-TPJ.csv** [contains information on the subjects]
      - **Area-TPJ_l_N10_nlin2ICBM152asym2009c_6.2_publicDOI_6aad6806f550fe95e2a780fce86ea6cc.nii.gz** [contains maps in the MNI ICBM 152 reference spaces, left hemisphere]
      - **Area-TPJ_l_N10_nlin2Stdcolin27_6.2_publicDOI_6aad6806f550fe95e2a780fce86ea6cc.nii.gz** [contains maps in the MNI Colin 27 reference spaces, left hemisphere]

Area-TPJ_r_N10_nlin2ICBM152asym2009c_6.2_publicDOI_71c8ebc47c3c8364a9c713ec51443218.nii.gz
[contains maps in the MNI ICBM 152 reference spaces, right hemisphere]

Area-TPJ_r_N10_nlin2Stdcolin27_6.2_publicDOI_71c8ebc47c3c8364a9c713ec51443218.nii.gz
[contains maps in the MNI Colin 27 reference space, right hemisphere]

CODE AVAILABILITY*

No code has been used for the delineations of the Area TPJ (STG/SMG) in the ten postmortem brains. Code regarding the computation of the Julich-Brain atlas can be retrieved from the original publication (Amunts et al., 2020).

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Author contributions

Daniel Zachlod mapped and analyzed the area. Hartmut Mohlberg calculated the probabilistic maps. Katrin Amunts developed the Julich-Brain atlas, supervised the acquisition and processing of data and contributed to the analysis.

REFERENCES

Amunts, K., Mohlberg, H., Bludau, S., and Zilles, K. (2020). Julich-Brain: A 3D probabilistic atlas of the human brain's cytoarchitecture. Science 369, 988–992. doi: 10.1126/science.abb4588

Evans, A. C., Janke, A. L., Collins, D. L., and Baillet, S. (2012). Brain templates and atlases. Neuroimage 62, 911–922. doi: 10.1016/j.neuroimage.2012.01.024